LAB SESSION:

Geiger-Müller counter properties and usage

FYS4505/9505 - FALL 2026

OBJECTIVE

The main purposes of this laboratory session are to learn how a Geiger-Müller counter works and how to use it in an experiment. There are two objectives: Firstly, the "plateau" and the operating voltage for a Geiger tube will be determined (Experiment 1); secondly, the half-lives of meta-stable Barium-137 ($^{137}_{56}\mathrm{Ba}^{m}$) and meta-stable Protactinium-234 ($^{234}_{91}\mathrm{Pa}^{m}$) will be measured (Experiment 2 & 3).

THEORY

The Geiger-Müller counter

The Geiger-Müller counter was invented in 1928 by H. Geiger and E.W. Müller. It is a tool for detecting the presence of radioactive particles that is designed to give a pulse signal independent of the type of particle and its energy as long as it has enough to ionize the gas in the detector.

A typical GM-counter is usually designed as a metallic cylindrical shape with a central wire going through the middle or axis of the cylindrical shape. The central wire or anode is kept at a positive potential and the metallic tube or cathode is grounded, the tube is usually filled with argon gas mixed with 5-10% of ethyl alcohol or halogens such as chlorine or bromine. One of the ends of the metallic tube will have a window which is the detection area, this is usually covered with a very thin material such as mica (it will very easily shatter upon contact so do NOT touch). There will be broken GM-counters available at the lab for inspection.

When ionizing radiation enters the detector it can ionize the molecules of the gas, creating what is called electron-ion pairs. The negatively charged free electrons will move towards the central wire whilst the positively charged ion will move towards the walls of the tube. The free electrons can now cause further ionization when accelerating towards the central wire, this can be repeated releasing

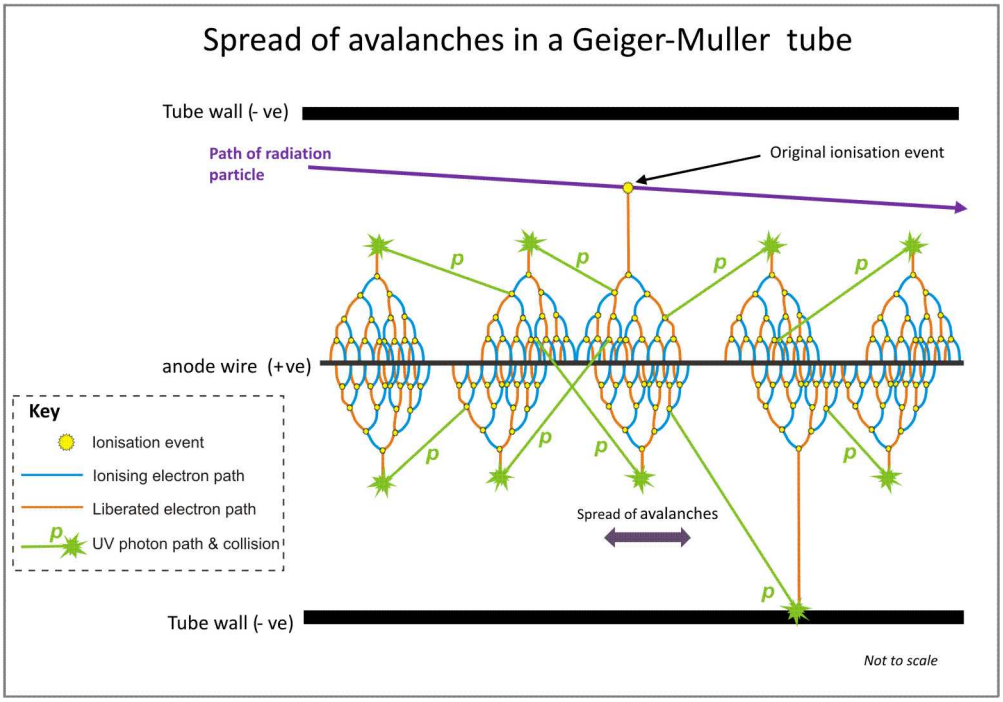


Figure 1: Illustration showing how avalanches originating from a single ionization event can spread throughout a Geiger-Müller tube. (Dougsim, 2012).

a cascade or local avalanche of electron-ion pairs. An illustration showing a possible scenario after a single ionization event in a GM-tube is shown in fig. 1. In addition to making electron-ion pairs the energy transferred by this effect will also excite gas molecules, which in turn will decay through emission of photons in the visible or ultraviolet region. These photons will travel some distance in the tube before photoelectric absorption will initiate another local avalanche of electron-ion pairs, repeating the whole process. This whole process happens in a matter of nanoseconds!

When enough avalanches reach the anode a drop in the potential between the anode and the cathode will give rise to a very large pulse. This pulse is communicated to an amplifier through a RC-circuit, reaching a scaler at the end. The amplitude of the pulse is independent of the type of radiation and it's energy, only telling us that there is ionizing radiation close to the GM-tube.

Radioactive decay and Half-life of a decaying nucleus

Radioactive decay is a random process. According to quantum theory there is no way to tell when a single nucleus will decay, however, what is predictable to some degree is the average or statistical properties of decaying nuclei. In other words, we know that the decay of nuclei will happen at a constant rate on average because the possibility of decay is there in the first place. Therefore some nuclei will decay within a short amount of time whilst others will stay in their unstable form for a much longer time: Translate this behaviour into a piece of material with billions of nuclei and

possibilities of decay times and you will have a continuous curve describing the lifetime. What you cannot describe is when 100% of the nuclei have decayed, that is why we use the term "half-life" instead. You can define a point in the curve where you know that approximately half of the nuclei in your piece of material have decayed and use this as a guideline. For example can you use this to look at the activity after two or three half-lives.

Naturally occurring radioactive nuclei undergo a combination of α, β , electron capture, and γ decays. Artificially produced nuclei may also decay by these processes plus by spontaneous fission, neutron emission and even proton and heavy-ion emission. The mathematical equation used to describe any of these decay process is called the "radioactive decay law" as seen in eq. (1) and illustrated in fig. 2. This function describes the number of decayed particles $N(t)$ after a time t of an initial number of nuclei N_0. It also states that the probability of decay per unit time is constant and independent of time. This "radioactive decay constant" is usually denoted λ and is unique for every single isotope.

$$N(t) = N_0 \exp(-\lambda t) \tag{1}$$

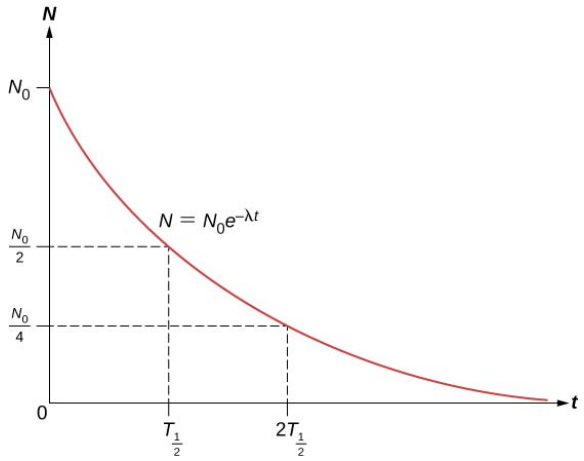


Figure 2: A graph showing the radioactive decay law as seen in eq. (1) with markers showing the half-life $T_{1/2}$ and half of the half-life $T_{1/4}$. (Samuel J. Ling (Truman State University) and authors, 2019)

A more common way of expressing the decay of a radioactive nucleus is its half-life. The half-life is defined as the time elapsed when the number of radioactive nuclei in a sample (i.e. the intensity of the radiation) is reduced to one half of its original value. It is related to the decay constant and given as:

$$t_{1/2} = \frac{0.693}{\lambda} \tag{2}$$

Where $t_{1/2}$ is the half-life. It is denoted in various forms such as $T_{1/2}$ or $\tau_{1/2}$.

Finally, the activity of a sample corresponds to the number of decays per unit time (rate) or change

in number of radioactive nuclei present. The activity depends on the number of nuclei present in the sample and given as:

$$A = \lambda N(t) = A_0 \exp(-\lambda t) \tag{3}$$

Units of activity:

1 Curie (Ci) = 3.7 $\times$ 10^10 disintegrations/s (dps)

1 Becquerel (Bq) = 1 disintegration/s

EXPERIMENTS

Experiment 1: Determining the plateau and operating voltage for GM-counter

Before we move on to explore different radioactive isotopes we have to set up the GM-counter. Firstly we need to determine the working voltage plateu for the GM-counter and find an optimal value for operation in the following experiments.

Procedure

1. Place the $^{137}_{55}$Cs source some distance from the GM-counter strapped in the stand, around 10cm should be a good placement. Take the yellow cap (or in some setups the black cap) off the GM-tube. Be careful NOT to touch the window end, this will damage the GM-tube.

2. Turn on the Scaler (Scaler 77) and the multimeter (FLUKE 8840A). The scaler provides the voltage to the GM-counter and also displays number of counts for each measurement. The FLUKE multimeter is used to read off the voltage precisely.

3. Start increasing the voltage from an initial value until the detector starts counting. Take the number of counts over a period of 10 seconds, thrice at each voltage value V. At the start and end of the plateau it is important to take smaller increments and more data points. Wait for a few seconds to let the voltage source stabilize itself. The range can be for example 300V to 700V.

Data Analysis

1. Plot the number of counts as a function of the voltage in Volt i.e., N(V) versus V.

2. Decide an optimal operating value for the voltage of the GM-counter. Explain the reasoning for your decision. Use this operating voltage for the next measurements.

Discussion for report

1. In fig. 3 below, classify different regions as a function of voltage as well as which type of decay each line represents.

2. Place the Geiger-Müller region in the plot and discuss your arguments for that.

3. Describe briefly the other possible regions in the plot.

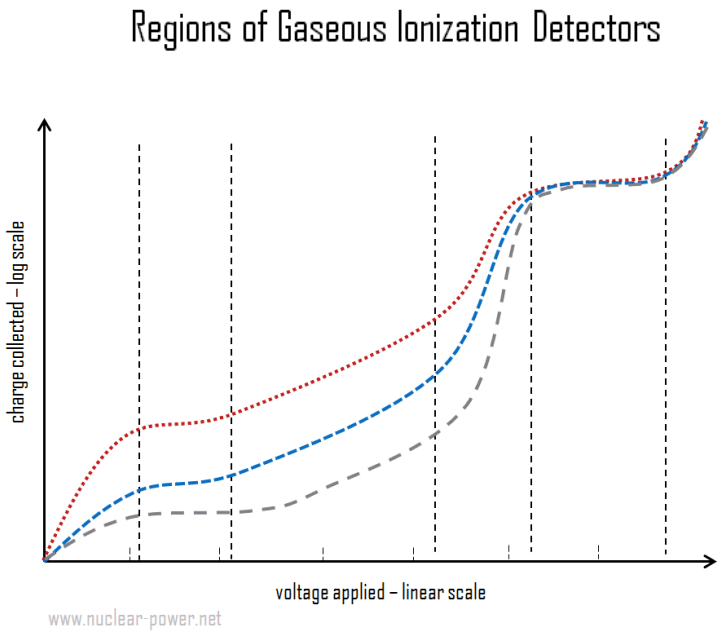


Figure 3: Illustration showing three possible outcomes for charge collected in gaseous ionization detectors. The three lines represent different types of radioactive decay.

Experiment 2: Half-life of metastable $^{137}_{56}Ba^{m}$

For the first part of the lab session we will look at a radionuclide generator producing $^{137}_{56}$Ba through beta-decay from $^{137}_{55}$Cs.

We will now try to determine the half-life of the metastable state experimentally, and compare it with the value found from a nuclear database. These steps need to be done within in a short amount of time, so be prepared!

Procedure

1. Scaler 77 should be set to "Manuell" counting, you will follow the time schedule in table 1 and read the number of counts N at each point t. Then push reset as the timer continues.

2. Start by filling the syringe with salt water.

3. Inject the salt water into the $^{137}_{55}$Cs container slowly, this will flush out any Barium-137 that has been produced. Fill a small container with the water and place it underneath the GM-counter.

4. One student can be responsible for checking the time. We will take measurement for 10 seconds and pause for 20 seconds in each time step, i.e., start the timer with 0 seconds and take measurement for 10 seconds and pause for 20 seconds. Reset the time interval. Start with the next 30 seconds again by taking 10 seconds of measurement and pausing for 20 seconds. Continue till the end of the time steps given in the table.

5. Remove the source and measure the count rate, i.e. background, for 100 seconds. Calculate average number of background counts for 10 seconds, "Counts$_{BG}$".

The measurement taking is now done, time to calculate and compare results. The next steps, Data Analysis and Discussion of the report will be done after the laboratory session in the coming days.

Data Analysis

1. Plot your data points as a function of t so that you have number of counts $N(t)$ on the y-xis and time t along the x-axis. Remember to account for the background counts. Do an exponential fitting of these points.

2. Plot the same plot with a logarithmic y-axis. Find the slope of this curve.

3. Use the information from the slope and calculate the half-life of $^{137}_{56}$Ba^m. Compare this value to what you found in a nuclear database.

Discussion for report

1. Why do we have to use half-life as a measurement of the lifetime of a decaying nucleus?

2. To find the half-life we need to use the radioactive decay law. Write it down and find out how we can calculate the half-life from this equation.

3. Go to nndc.bnl.gov or your favorite nuclear database and use it to find the decay information of $^{137}_{55}$Cs. Find the decay scheme for $^{137}_{55}$Cs $\rightarrow$ $^{137}_{56}$Ba and the metastable state we expect to see signals from, what is the decay energy and lifetime of the state? What is the probability of seeing signals from other states in $^{137}_{56}$Ba?

4. Discuss in detail the meaning of the slope you found in "Data Analysis" task number 2.

5. Does the half-life of a radioisotope change over time?

6. Describe all the possible error sources you can think of that might have influenced your results.

t (sec)	Counts	Log.(Counts-Counts$_{BG}$))	t (sec)	Counts	Log.(Counts-Counts$_{BG}$))
0			510		
30			540		
60			570		
90			600		
120			630		
150			660		
180			690		
210			720		
240			750		
270			780		
300			810		
330			840		
360			870		
390			900		
420			930		
450			960		
480			990		

Table 1: Data sheet for data points when calculating the half-life.

Experiment 3: Measuring the half-life for metastable $^{234}_{91}Pa^{m}$

We will now move on to another radionuclide generator. This one produces a metastable state of $^{234}_{91}Pa^{m}$ through a famous decay chain still active on our earth today. The container is filled with an acidic solution with the mother nuclide and an organic solution plus a water-immiscible organic liquid which will extract $^{234}_{91}Pa^{m}$ when the container is shaken. The two liquids will separate, with the top clear layer being filled with the organic liquid and $^{234}_{91}Pa^{m}$.

Procedure

1. Scaler 77 should be set to "Manuell" counting, you will follow the time schedule in table table 2 and read off the number of counts N at each point t. Then push reset as the timer continues.

t (sec)	Counts	Log.(Counts-Counts$_{BG}$)	t (sec)	Counts	Log.(Counts-Counts$_{BG}$)
20			280		
40			300		
60			320		
80			340		
100			360		
120			380		
140			400		
160			420		
180			440		
200			460		
220			480		
240			500		
260			520		

Table 2: Data sheet for data points when calculating the half-life.

2. First find a good position for the GM-counter so that there is just room (no contact) for the generator below.

3. With the generator in place, **but before** shaking the generator, measure the background for 60 seconds 2-3 times.

4. Carefully remove and then shake the container as hard as possible for 10-15 seconds. Put the container underneath the GM-counter. Either visually confirm that the two liquids have separated or take some measurements (not critical to record time or counts) and wait for the counts/measurement to start a downward trend.

5. Start the measurement taking. Record the number of counts per 10 s every 20 seconds for 7-8 minutes.

6. Leave the generator in place and carry out an additional background measurement.

Data Analysis

1. Plot your data points as a function of t so that you have number of counts N on the y-xis and time t along the x-axis. Remember to account for the background, correctly taking into account the time intervals of the background and decay measurements. Do an exponential fitting of these points.

2. Plot the logarithm of the background-subtracted data and fit these results with a straight line.

3. Use the information from the slope and calculate the half-life of $^{234}_{91}\mathrm{Pa}^{m}$. Compare this value to what you found in a nuclear database.

Discussion for report

1. Which famous decay chain is $^{234}_{91}\mathrm{Pa}^{m}$ produced through? What was the mother nuclide in the radionuclide generator?

2. When this radionuclide generator was produced, which nuclide do you think was it filled with?

3. Find the decay path from the mother nuclide to $^{234}_{91}\mathrm{Pa}^{m}$ and identify the metastable state and the half-life of this state. What are the other options of decay from this metastable state?

References

Dougsim, Wikipedia (2012). *Spread of avalanches in G-M tube*. URL: https://commons.wikimedia.org/wiki/File:Spread_of_avalanches_in_G-M_tube.jpg (visited on 07/08/2019).

Samuel J. Ling (Truman State University), Jeff Sanny (Loyola Marymount University) and Bill Moebs with many contributing authors (2019). *Plot of radioactive decay law*. URL: https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_(OpenStax)/Map%3A_University_Physics_III_-_Optics_and_Modern_Physics_(OpenStax)/10%3A__Nuclear_Physics/10.3%3A_Radioactive_Decay (visited on 20/08/2019).